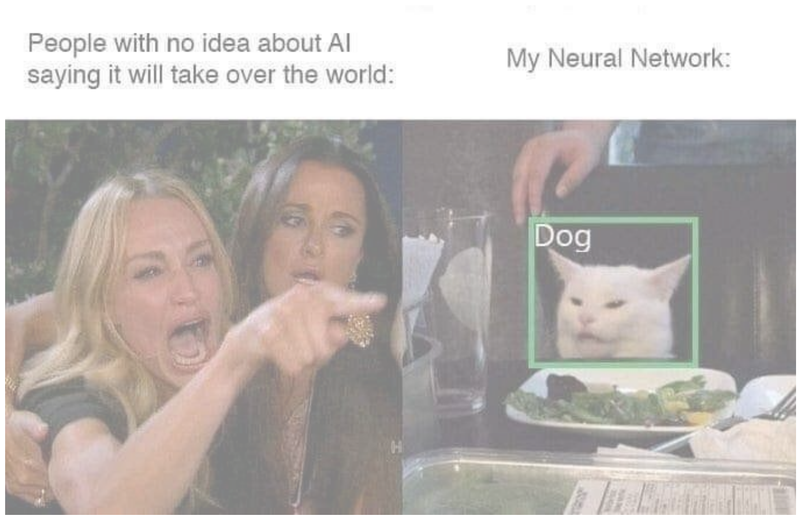

OptiML Lab Workshop (2026.02.13)

Unraveling & Overcoming Challenges in ML:
Generalizability, Adaptability, & Multifacetedness

Hanseul Cho
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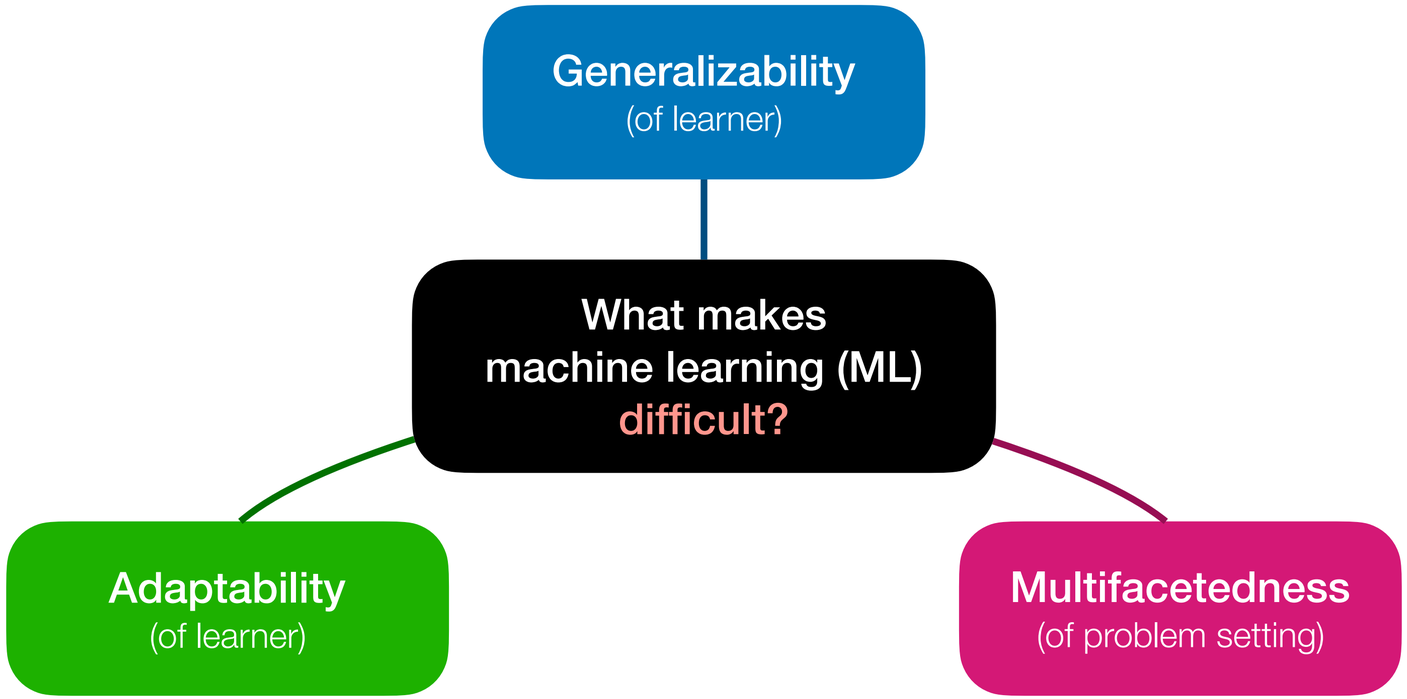
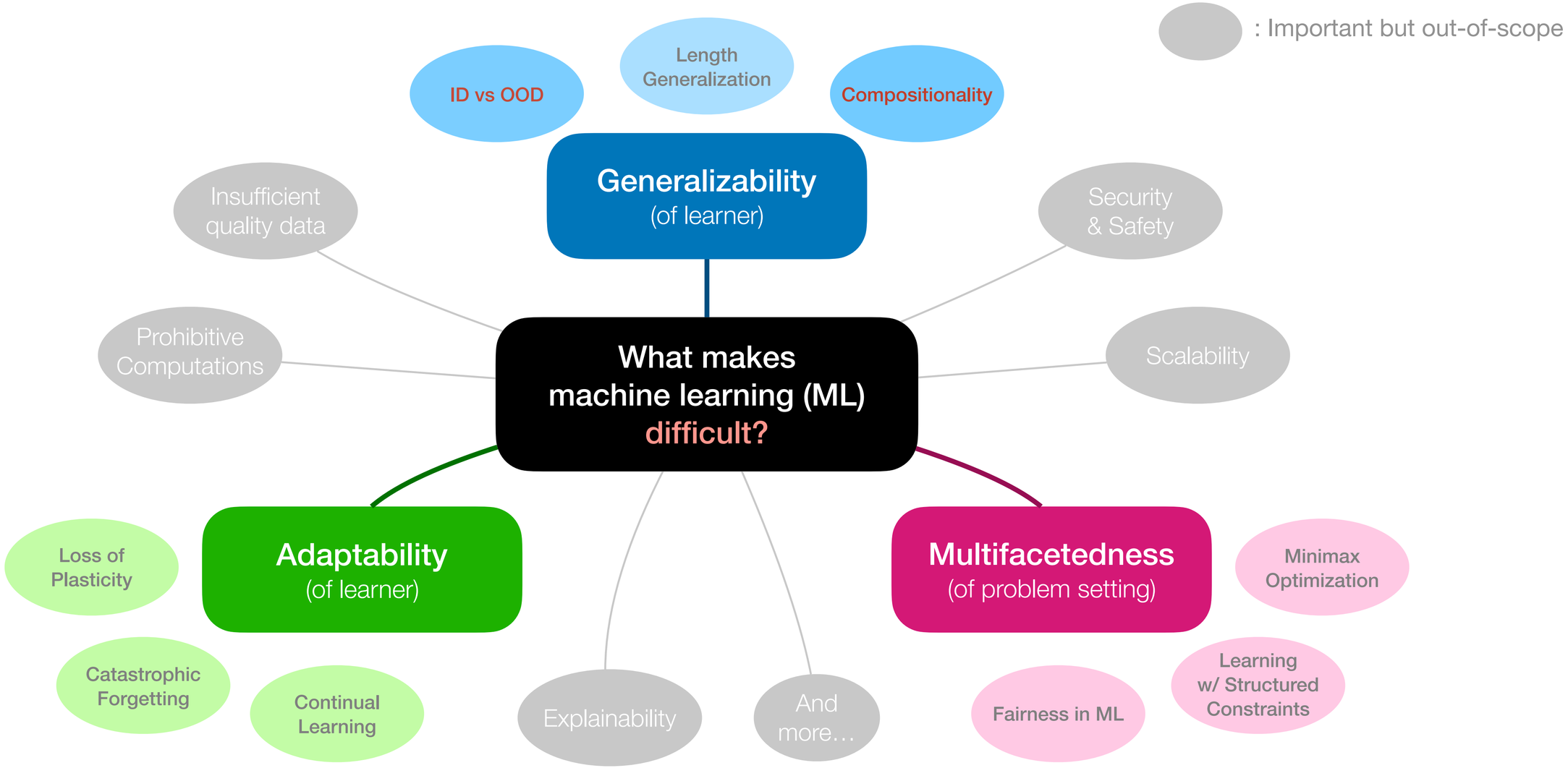
OptiML Optimization & Machine Learning Laboratory

My Research Interests

"*Deep Learning Magic* is NOT a Piece of 🍰."



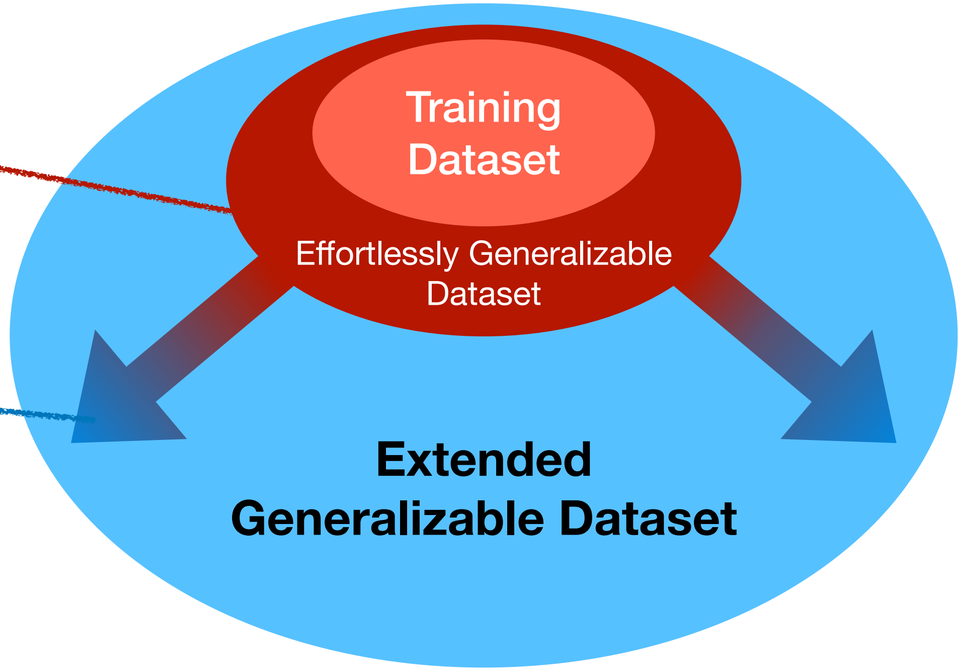
My Research Interests



Generalizability

Generalization of (Sequence-to-Sequence) Models

- [Theory:] How far can a model generalize outside of the training dataset?
- [Engineering:] How can we expand the scope of the model's generalization?

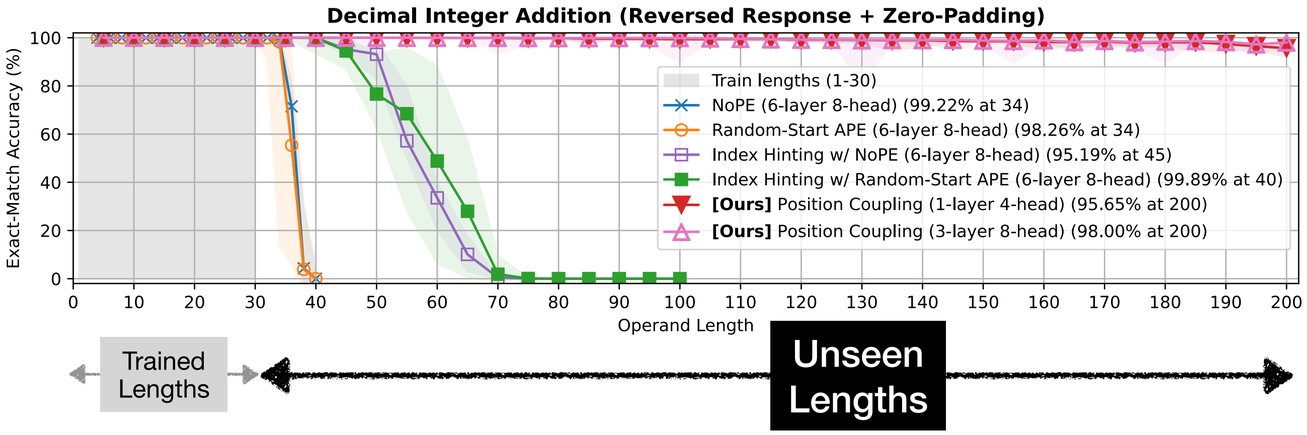


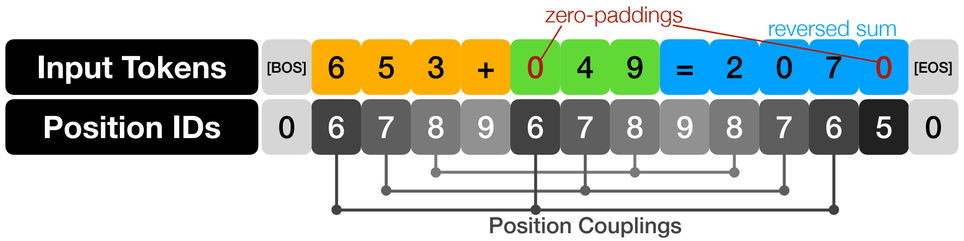
Generalizability

Length Generalization: Train Short, Test Long

- Position Coupling [CCA+24; CCBY25] (w/ Jaeyoung & Google)
  - Enhances the length generalization of Transformers* in many **arithmetic/algorithmic** tasks.

    * Transformers with causal masks; Auto-regressive; Decoder-only.



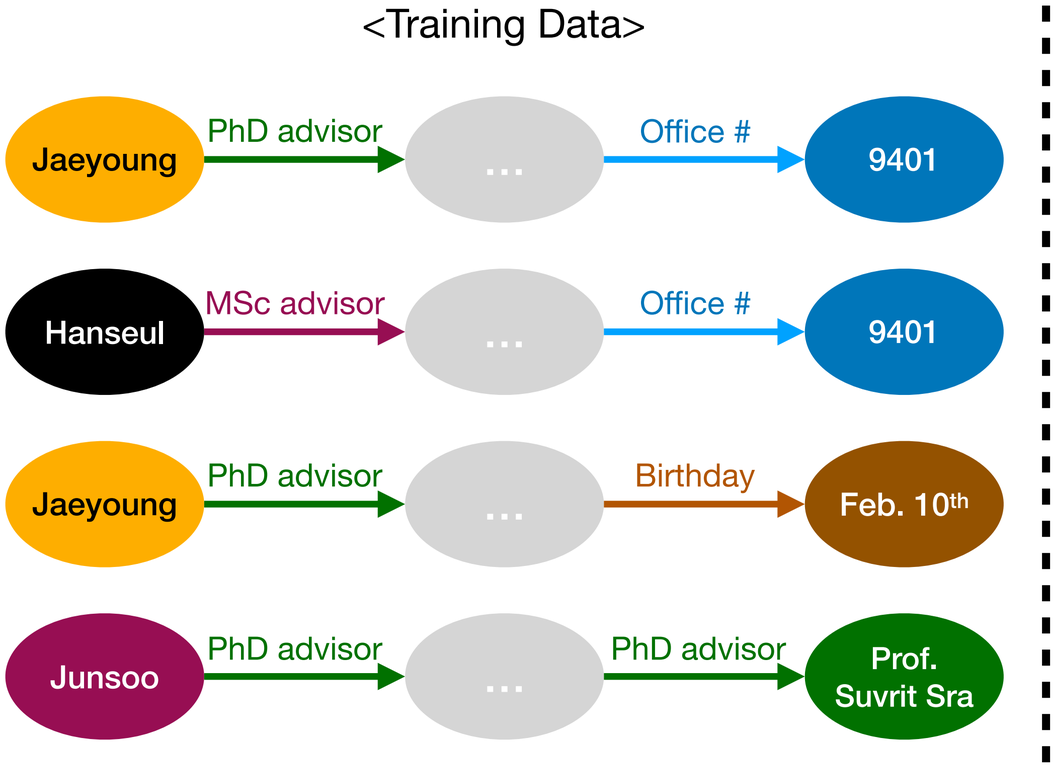


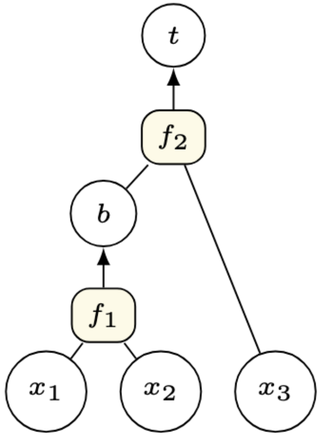
⬆️ Illustration of Position Coupling for the long integer addition task.

Generalizability

Compositional Generalization: Re-assemble Knowledge

- Two-hop composition task



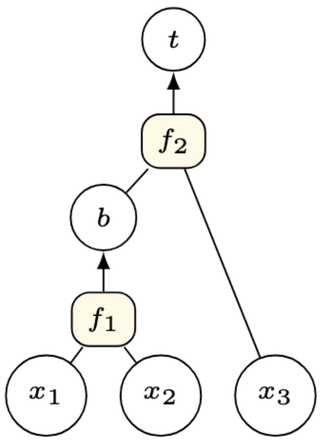


(a) 2-Hop

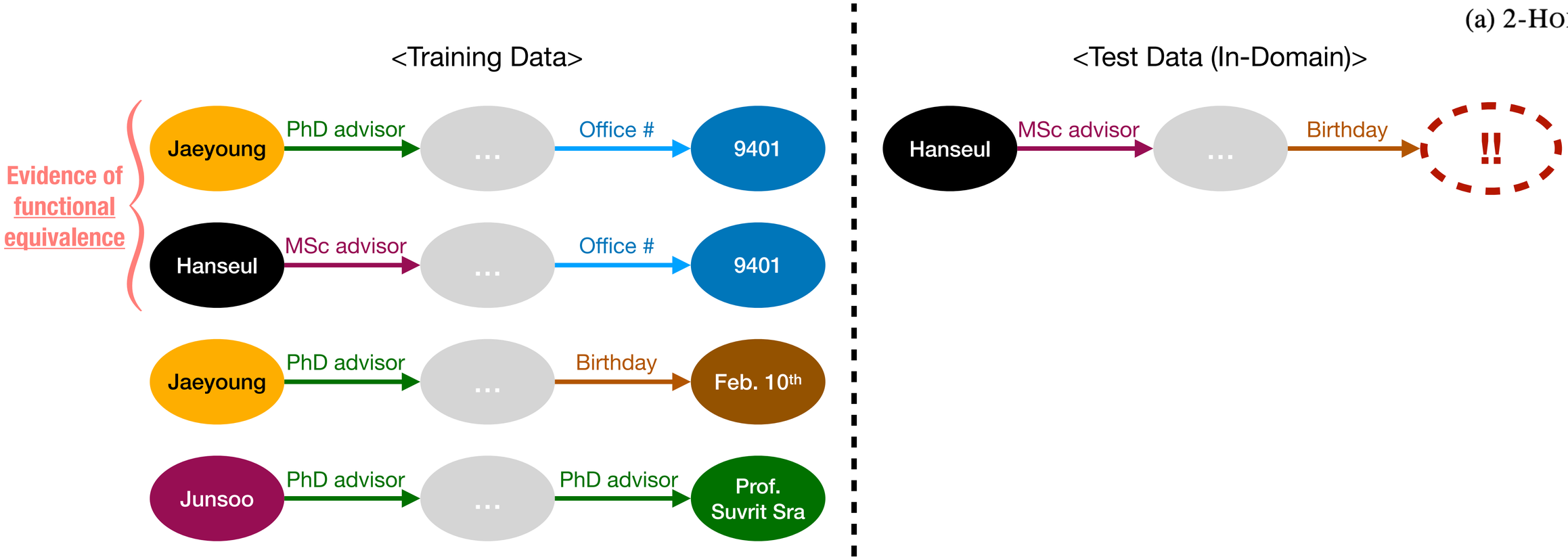
Generalizability

Compositional Generalization: Re-assemble Knowledge

- Two-hop composition task



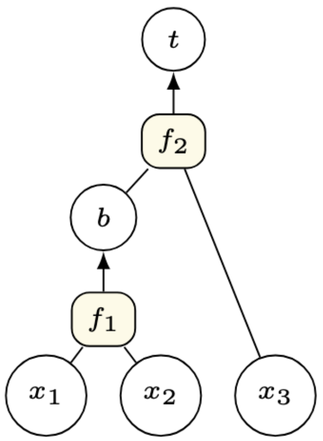
(a) 2-Hop



Generalizability

Compositional Generalization: Re-assemble Knowledge

- Two-hop composition task



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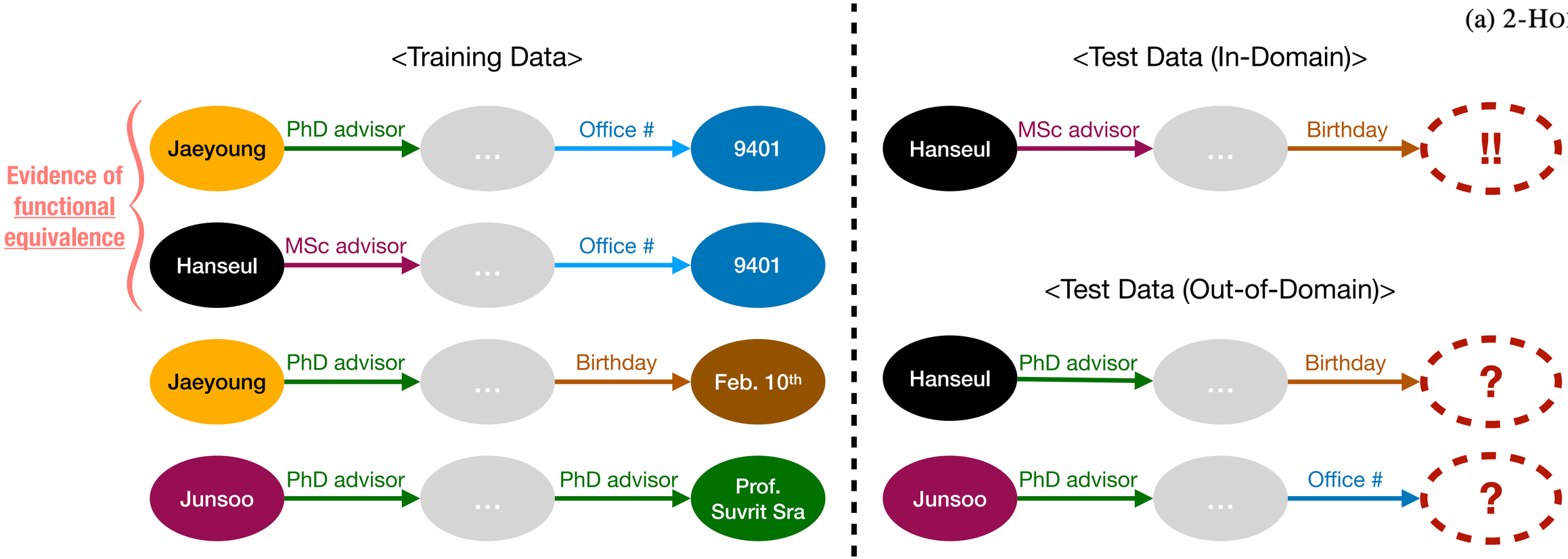
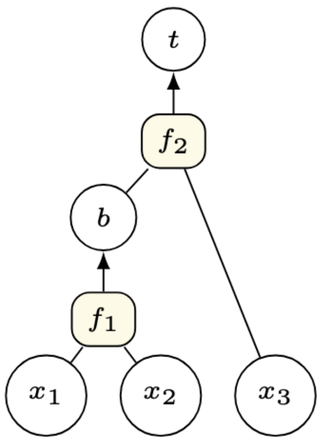


Generalizability

Compositional Generalization: Re-assemble Knowledge



(a) 2-Hop

- Theoretical Framework: "**Coverage**" of two-hop composition [CPC+26]
  - **Functional k-equivalence:**

    assume: universal constant

    k distinct evidence

    ➡ (e.g.,) (x_1, x_2) & $(\check{x}_1, \check{x}_2)$ are *functionally k-equivalent* at $\{1,2\}$ if there are $\boxed{x_3^{(1)}, \ldots, x_3^{(k)}}$ such that

$$F\left(x_1, x_2, x_3^{(j)}\right) = F\left(\check{x}_1, \check{x}_2, x_3^{(j)}\right) \quad (j = 1, \cdots, k).$$

  - **k-Coverage:**

    ➡ For a train dataset $D \subseteq X_1 \times X_2 \times X_3$, the set $\mathrm{Cover}_k(D) \subseteq X_1 \times X_2 \times X_3$ is the largest set of input sequences whose labels can be inferred solely by the functional k-equivalences.

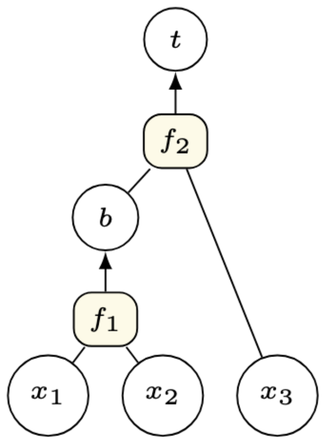
  - **In-domain Closure**:

    ➡ For a train dataset $D \subseteq X_1 \times X_2 \times X_3$, the set $\overline{D} \subseteq X_1 \times X_2 \times X_3$ is the largest set of input sequences whose labels can be exactly obtained using the ground-truth intermediate calculations (f_1, f_2) on subsequences in D.

★ Question: **Can the k-Coverage cover the whole in-domain closure, i.e., $\overline{D} \subseteq \mathrm{Cover}_k(D)$?**

Generalizability

Compositional Generalization: Re-assemble Knowledge



(a) 2-Hop

- Implication in **data scaling law** for two-hop tasks [CPC+26]
  - Ground-truth function $F(x_1, x_2, x_3) = f_2(f_1(x_1, x_2), x_3)$ $(f_1 : X_1 \times X_2 \to B := f_1(X_1 \times X_2);\ f_2 : B \times X_3 \to Y)$
  - Vocab size $|X_1| = |X_2| = |X_3| = |B| = n$, Randomly (with replacement) sampled train dataset D of size $N \gtrsim n^2$
  - We show that $N = \widetilde{\Theta}\left(n^{2.5-\frac{0.5}{k}}\right)$ is sufficient (and necessary if $k \geq 2$) to guarantee $\overline{D} \subseteq \mathrm{Cover}_k(D)$ w.h.p (Thm 6.1)!

Theorem F.7 ($k \geq 2$, Simplified UB) [CPC+26]

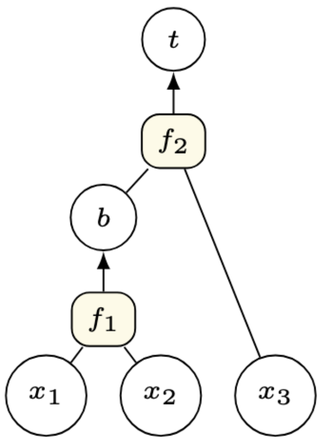
Consider a learner that generalizes within its $\mathrm{Cover}_k(D)$. Let $\hat{n} = \min_{b \in B} |f_1^{-1}(b)|$, $\check{n} = \max_{b \in B} |f_1^{-1}(b)|$.

Assume $|X_3| = \Omega(\check{n})$. Fix any $\delta > 0$. Then, there exists $N_\delta \leq O\left(\max\left\{|X_1 \times X_2| \cdot \sqrt{|X_3|} \cdot \left(\frac{k!\,\log \hat{n}}{\hat{n}}\right)^{\frac{1}{2k}}\right\}, \log\left(\frac{1}{\delta}\right)\right)$

such that: for any large $\hat{n}$ and any $N > N_\delta$, the learner achieves $\overline{D} \subseteq \mathrm{Cover}_k(D)$ with probability $\geq 1 - \delta$.

Generalizability

Compositional Generalization: Re-assemble Knowledge



(a) 2-Hop

- Implication in **data scaling law** for two-hop tasks [CPC$^+$26]
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Theorem F.16 ($k \geq 2$, Simplified LB) [CPC$^+$26]

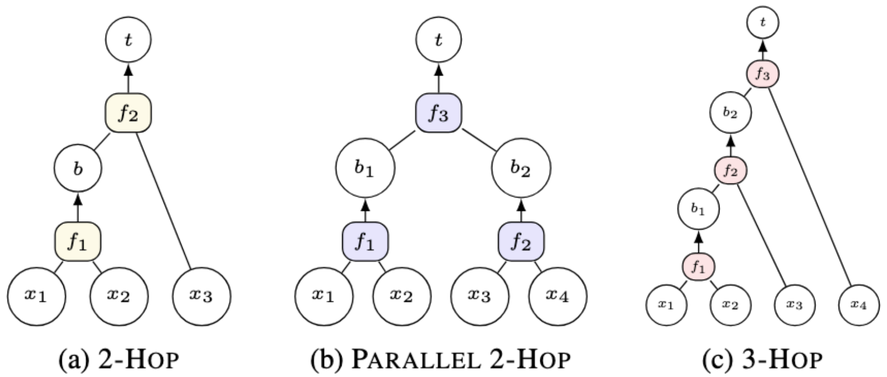
Consider a learner that generalizes within its $\mathrm{Cover}_k(D)$. Let $\hat{n} = \min_{b \in B} |f_1^{-1}(b)|$, $\check{n} = \max_{b \in B} |f_1^{-1}(b)|$.

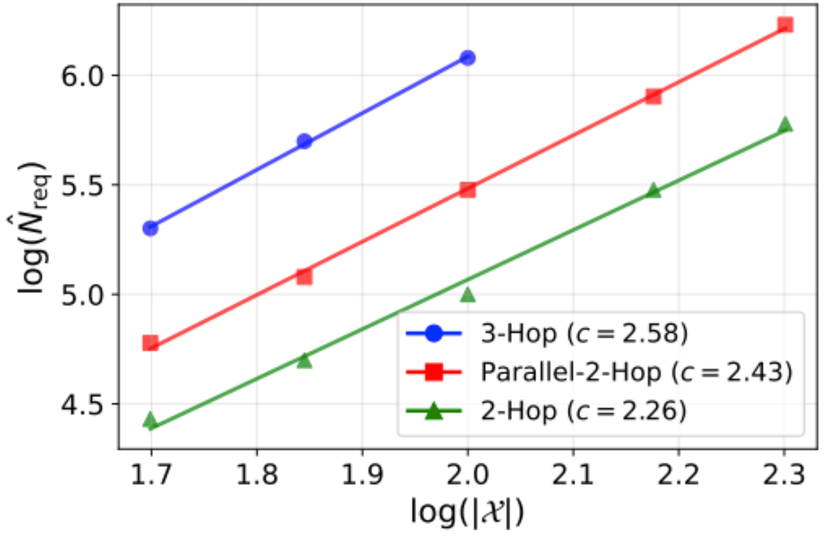
Assume $|X_3| = \Theta(\hat{n})$. Fix any $\delta > 0$. Then, there exists $\widetilde{N}_\delta \geq |X_1 \times X_2| \cdot \sqrt{|X_3|} \cdot \left(\frac{(k-1)!\ \log \hat{n}}{\hat{n}}\right)^{\frac{1}{2k}}$ such that:

for any large $\hat{n}$ and any integer N satisfying $|X_1 \times X_2| \cdot \max\left\{\log \frac{4|X_1 \times X_2|}{\delta}, \frac{|X_3|}{\hat{n}} \log \frac{4|B \times X_3|}{\delta}\right\} \leq N < \widetilde{N}_\delta$, the learner must suffer $\overline{D} \not\subseteq \mathrm{Cover}_k(D)$ with probability $\geq 1 - \delta$.

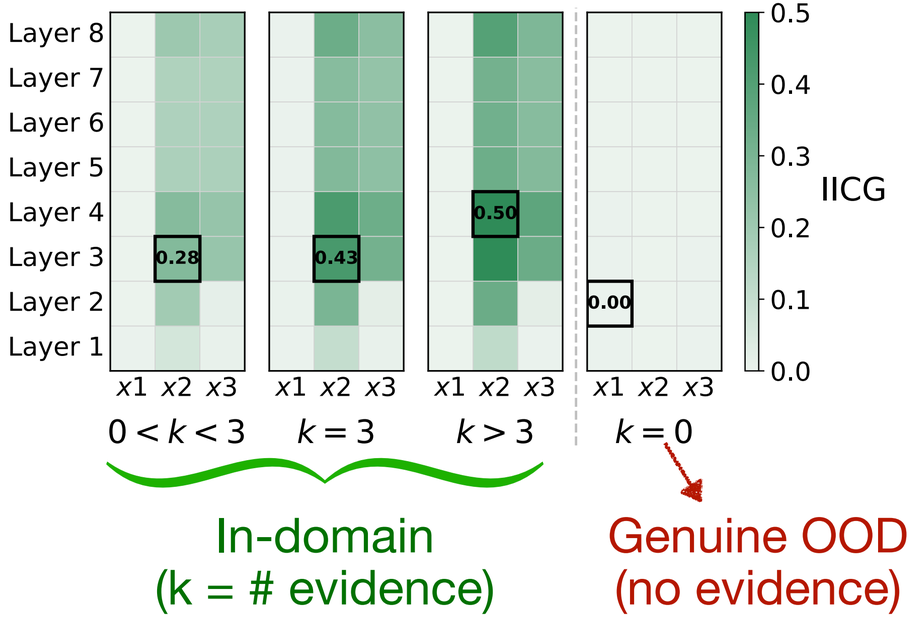
Generalizability

Compositional Generalization: Re-assemble Knowledge





*Intra-Inter Cosine Gap: $\text{IICG} = \overline{\cos}_{\text{intra}} - \overline{\cos}_{\text{inter}}$



Adaptability*

Towards an Adaptable Learner under an Evolving Environment

- The real world is far from the i.i.d. assumption.
- Every learner with finite capacity under changing circumstances should keep a balance between:

Stability (to remember old things)

⬍

Plasticity (to adapt to new things)

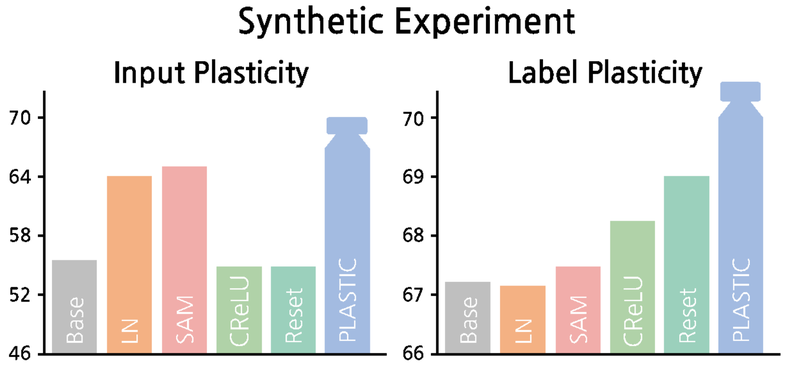
* Adapt**ability**: the capability to **change one's behavior** to fit new circumstances. (An adapt**able** model can **change its inference/thinking** w.r.t surroundings)

* Adapt**ivity**: the capability to **change oneself** to fit new circumstances. (An adapt**ive** model can **change its shape/architecture** w.r.t the surroundings)

Adaptability

Loss of Plasticity in Reinforcement/Incremental Learning & “Re-Learning”

- “PLASTIC [LCK+23]” (w/ DAVIAN lab & OSI lab)
  - SAM optim. & LayerNorm ➡️ Input-plasticity (w.r.t $p(x)$) 📈
  - Periodic Reset & CReLU ➡️ Label-plasticity (w.r.t $p(y \mid x)$) 📈
  - Combine ➡️ Synergetic improvement for sample-efficient RL!



- “DASH [SOCY24]” (with Baekrok & Junsoo)
  - Intuition: selectively & periodically forget the overfitted noises.

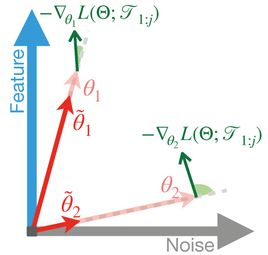


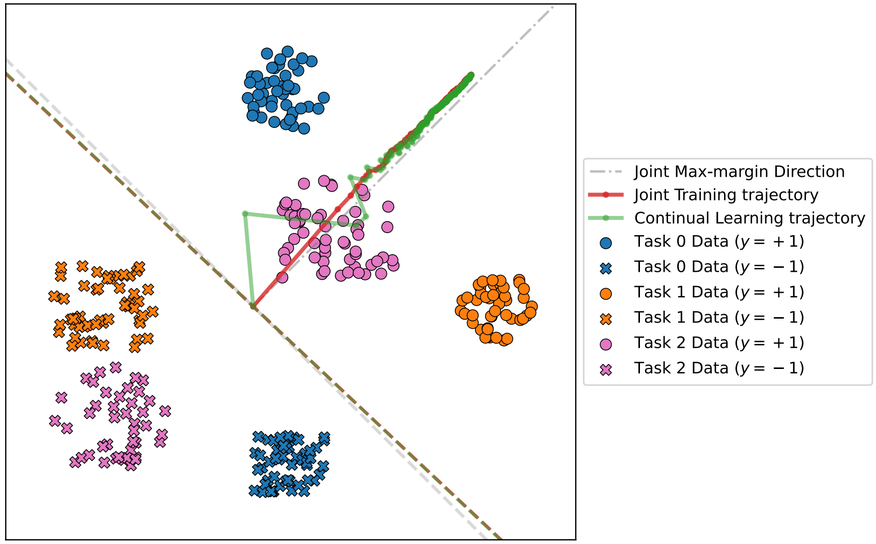
Figure 4: Illustration of DASH. We compute the loss L with training data $\mathcal{T}_{1:j}$ and obtain the negative gradient. Then, we shrink the weights proportionally to the cosine similarity between the current weight θ and $\nabla_\theta L$, resulting in $\tilde{\theta}$.

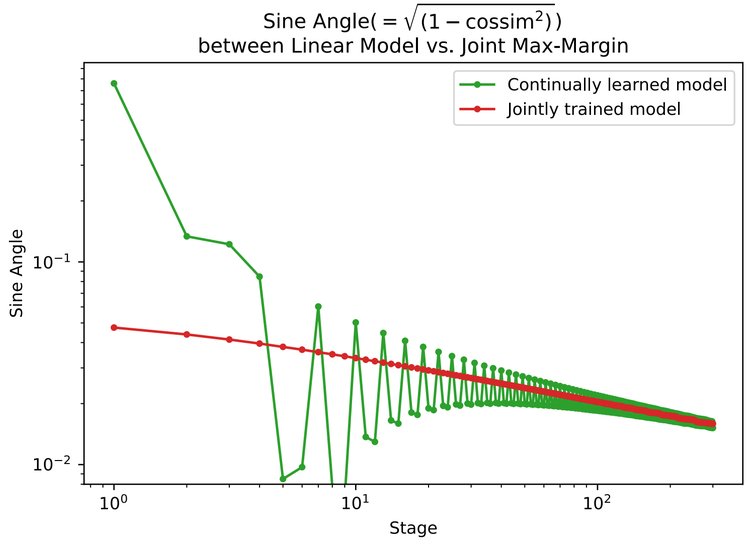
- Still unclear:
  1. Why does weight-space smoothness (e.g., via SAM) encourage input-robustness? (Or does it?)
  2. Why are the (periodic) re-learning methods (reset, S&P…) so effective for maintaining plasticity?

Adaptability

Theoretical Analysis of Continual Learning for Classifications

- [JCY25] study implicit bias & convergence of GD in **continual** logistic regression (with Hyunji, a former intern)
- Part 1: Jointly (linearly) separable tasks
  - GD (with small LR) directionally converges to the joint max-margin direction
  - The convergence behavior of the per-cycle forgetting (towards zero) depends on the data alignment btw tasks
- Part 2: Jointly non-separable tasks
  - GD (with small LR) converges towards a joint minimum of total train loss





Multifacetedness

Learning Problems with Multiple (Conflicting) Goals

- NOT every learning problem falls under the ☂ of a simple minimization ($\min_x f(x)$).
- Plenty of meaningful setups consider multi-dimensional objectives and constraints.
  - The multiple goals might conflict, leading to a trade-off.
  - Handling orthogonal goals is also difficult: what should we prioritize?

Multifacetedness, *n.*

1. *literal*. The state or condition of having many faces or facets.
2. *figurative*. The state or condition of having many aspects, skills, or qualities.

Multifacetedness

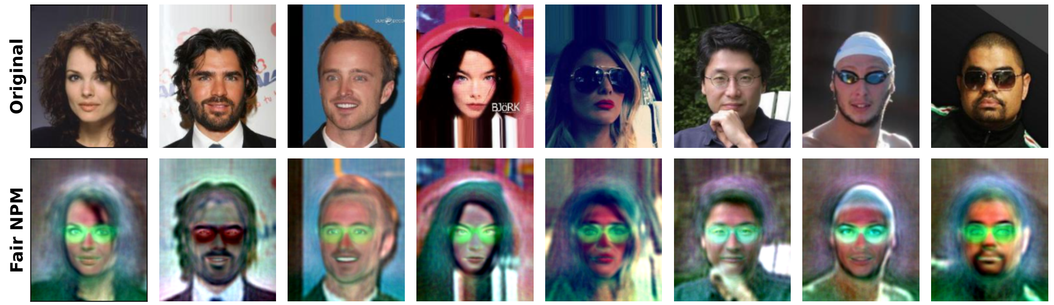
Minimax Optimization

- "Minimizing the (Partial) Maxima": $\min_x \max_y f(x, y)$
- Many open research directions.
  - Importing methods from minimization to minimax optimization
    - **Shuffling** accelerates stochastic minimax optimization [CY23]
  - Convergence analysis of minimax algorithms & designing faster algorithms.
    - **`Alt-GDA`** is *faster*† than **`Sim-GDA`**; + we designed a *faster*† algorithm (**`Alex-GDA`**) [LCY24]. (w/ Jaewook)
      - The optimal configuration (i.e., algorithm-specific hyperparameter) for **`Alex-GDA`** is unknown.
  - More extensions: bi-level optimization, N-player games, …

†"Faster" under strongly-convex-strongly-concave Lipschitz smooth objectives + deterministic (i.e., non-stochastic) setting.

Multifacetedness

Fairness in ML



Sensitive Attribute: Eyeglasses (Left four: False, Right four: True)

- Fairness can be considered in ML in various formats
  - Constrained optimization, bi-level optimization, multi-objective learning…
  - In the era of over-parametrized models, which have multiple (local) optima, I hope I can eventually study algorithms with an **implicit/explicit bias toward the fairest solution among the set of optima**. Also, I believe any other social aspects of ML can be discussed in this way.

- **Fair Streaming PCA** [LCYY23] (work with Nick & OSI lab)
  - Dimensionality reduction algorithm w/ (1) large enough explained variance + (2) representational fairness + (3) memory efficiency (+ Statistical (PAFO) guarantee!)
  - We defined an amenable form of second-order representational fairness as
    - Eliminating the mean difference & Eliminating the dominant eigenspace of the covariance difference
  - Ongoing work: Better notion of second-order representational fairness + Suitable algorithm

References

1) [**C**Y23] **Hanseul Cho** and Chulhee Yun. SGDA with shuffling: faster convergence for nonconvex-PL minimax optimization. ICLR 2023.
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3) [L**C**K+23] Hojoon Lee*, **Hanseul Cho***, Hyunseung Kim*, Daehoon Gwak, Joonkee Kim, Jaegul Choo, Se-Young Yun, and Chulhee Yun. “PLASTIC: Improving Input and Label Plasticity for Sample Efficient Reinforcement Learning.” NeurIPS 2023.
4) [L**C**Y24] Jaewook Lee*, **Hanseul Cho***, Chulhee Yun. “Fundamental Benefit of Alternating Updates in Minimax Optimization.” ICML 2024.
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6) [SO**C**Y24] Baekrok Shin*, Junsoo Oh*, **Hanseul Cho**, and Chulhee Yun. “DASH: Warm-starting Neural Network Training in Stationary Settings without Loss of Plasticity.” NeurIPS 2024.
7) [**C**CBY25] **Hanseul Cho***, Jaeyoung Cha*, Srinadh Bhojanapalli, and Chulhee Yun. “Arithmetic Transformers can Length-Generalize in Both Operand Length and Count.” ICLR 2025.
8) [J**C**Y25] Hyunji Jung*, **Hanseul Cho***, and Chulhee Yun. “Convergence and implicit bias of gradient descent on continual linear classification.” ICLR 2025.
9) [CP**C**+26] Hoyeon Chang*, Jinho Park*, **Hanseul Cho***, Sohee Yang, Miyoung Ko, Hyeonbin Hwang, Seungpil Won, Dohaeng Lee, Youbin Ahn, Minjoon Seo. “Characterizing Pattern Matching and Its Limits on Compositional Task Structures.” ICLR 2026. (NEW!)

For those interested:

- Visit my GitHub blog: `hanseuljo.github.io`

Feel free to reach out for questions, discussions, and collaborations! 🤗